

Probability for UPSC ISS 2027

The Top-1% Trap Compendium

Paradoxes · Counterexamples · Examiner Ambush Points · Expected MCQ Angles

Statistics Paper–I (Objective) — Complete Probability Syllabus Coverage

Compiled in the style of a senior ISS faculty

Edition: July 2026 — built for the 2027 attempt

How to use this document. Every pointer below is a place where an MCQ can be — and historically has been — set to separate the top 1% from the rest. Read a pointer, close the file, and reconstruct *why* the trap works. If you cannot, you do not own it yet. Statistics–I is 80 questions in 120 minutes (200 marks, 2.5 marks per question, $-1/3$ of the question’s marks per wrong answer). The paper is not testing whether you can integrate; it is testing whether you can *refuse to be fooled in 90 seconds*. Probability and its limit theory typically drive a large, score-deciding block of Paper–I.

Contents

1	The Unconventional Path (Read This First)	2
2	Classical and Axiomatic Definitions — Where MCQs Hide	2
2.1	Classical (Laplace) definition: the four attack points	2
2.2	Kolmogorov’s axioms and their exact strength	2
2.3	Independence: the single richest MCQ mine in foundations	3
3	Conditional Probability, Total Probability, Bayes — The Paradox Belt	3
3.1	Machinery with trap annotations	3
3.2	The paradox arsenal (each is a ready-made 2027 MCQ)	4
4	Random Variables and Distribution Functions — Fine Print That Becomes MCQs	5
4.1	Distribution function (CDF): the exact properties	5
4.2	Building new CDFs from old — a favourite “which of these is a CDF” battery . . .	6
4.3	Random variables: definitional traps	6
5	Standard Distributions — Distribution-by-Distribution Ambush Points	7
5.1	Bernoulli(p)	7
5.2	Binomial(n, p)	7
5.3	Poisson(λ)	7
5.4	Geometric(p) — the discrete memoryless law	8

5.5	Uniform / Rectangular	8
5.6	Exponential(λ) — the continuous memoryless law	8
5.7	Normal(μ, σ^2)	9
5.8	Cauchy(θ, σ) — the professional counterexample	9
5.9	Hypergeometric(N, K, n)	9
5.10	Multinomial($n; p_1, \dots, p_k$)	10
5.11	Laplace (double exponential)(μ, b)	10
5.12	Negative Binomial(r, p)	10
5.13	Beta(α, β)	10
5.14	Gamma(α, λ) (shape–rate)	11
5.15	Lognormal(μ, σ^2)	11
5.16	Cross-family cheat tables (matching-type questions)	11
6	Random Vectors: Joint, Marginal, Conditional — and Transformations	12
6.1	Joint vs marginal: the one-way street	12
6.2	Independence of random variables: precision points	12
6.3	Distributions of functions of random variables	12
7	Modes of Convergence — The Counterexample Arsenal	13
7.1	The hierarchy (memorise as a picture)	13
7.2	Definitions with the exact quantifier traps	13
7.3	The four counterexamples that answer 90% of convergence MCQs	13
7.4	Theorems the examiner expects you to wield in one line	14
8	Expectation and Conditional Expectation	14
8.1	Existence and computation traps	14
8.2	Conditional expectation: the seven laws	15
9	Generating Functions: CF, MGF, PGF — Inversion, Uniqueness, Continuity	15
9.1	Characteristic functions: what is always true	15
9.2	The three big theorems (statement precision wins the mark)	16
9.3	MGF and PGF: existence fine print	16
10	Zero–One Laws and the Classical Inequalities	16
10.1	Borel–Cantelli and Borel’s 0–1 law	16
10.2	Kolmogorov’s 0–1 law	17
10.3	Tchebycheff, Markov, Kolmogorov: exact hypotheses, exact bounds	17
11	Laws of Large Numbers and Central Limit Theorems	17
11.1	WLLN vs SLLN: hypothesis bookkeeping (pure MCQ fuel)	17
11.2	CLT: the three named versions	18
12	The 2027 Expected-Question Bank: 60 Trap Statements with Verdicts	19
12.1	Foundations and conditioning	19

12.2 Random variables, CDFs, distributions	19
12.3 Convergence, expectation, generating functions, limit theorems	21
13 The Mistakes That Waste 80% of Beginners' Time	22
14 The Aggressive 2027 Practice Plan	23
14.1 Phase 0 — Calibration (Weeks 1–2)	23
14.2 Phase 1 — Trap-first mastery (Weeks 3–14; the engine room)	23
14.3 Phase 2 — Mixed-speed training (Weeks 15–30)	23
14.4 Phase 3 — Exam simulation (Weeks 31–42)	24
14.5 Phase 4 — Peak (final 4 weeks before the paper)	24
14.6 Non-negotiables (the whole plan in five lines)	24

1 The Unconventional Path (Read This First)

KEY. The traditional path is: theory $\rightarrow$ solved examples $\rightarrow$ exercises $\rightarrow$ mocks. It takes 14–18 months and produces students who “know” probability but still lose 15–20 questions to traps. The inverted path used by toppers:

1. **Traps first.** Learn every statement in this compendium as a *true/false object*, with the counterexample attached. MCQs are built from exactly these fault lines.
2. **PYQs as the syllabus.** Solve ISS Statistics–I previous-year questions (2016–2025) for probability *before* finishing theory. Also mine IIT JAM (MS), GATE (ST), and CSIR–NET Part B/C for question patterns: these three exams share question-setters’ culture with ISS.
3. **Counterexample flashcards.** Ninety percent of “assertion–reason” and “which of the following is/are correct” items die instantly if you carry a mental deck of ~ 40 counterexamples (all included below, marked **TRAP.**).
4. **Error ledger.** Every mistake goes into a ledger with the trap-class it belongs to. You revise *the ledger*, not the textbook.
5. **Speed arithmetic of guessing.** With 4 options, $+2.5$ for right, -0.833 for wrong: random guessing has expected value $0.25(2.5) - 0.75(0.833) = 0$ (neutral); eliminating even *one* option makes guessing strictly profitable: $\frac{1}{3}(2.5) - \frac{2}{3}(0.833) \approx +0.28$. Top scorers never leave a question blank once one option is eliminated.

2 Classical and Axiomatic Definitions — Where MCQs Hide

2.1 Classical (Laplace) definition: the four attack points

1. $P(A) = \frac{\text{favourable cases}}{\text{total cases}}$ *only if* outcomes are finite **and** equally likely. **MCQ angle:** “Probability that a natural number chosen at random is even” — ill-posed classically (infinite sample space); needs limiting (natural density) arguments.
2. **TRAP.** The definition is **circular**: “equally likely” already means “equally probable”. A favourite one-liner for assertion–reason items.
3. **TRAP.** It cannot handle unequally likely outcomes (biased coin) or continuous sample spaces. Geometric probability (favourable measure/total measure) extends it, but then Bertrand’s chord paradox shows the extension is *not unique* (§5).
4. Classical probability is automatically a valid probability measure (it satisfies Kolmogorov’s axioms); the converse direction is what fails.

2.2 Kolmogorov’s axioms and their exact strength

On a sample space Ω with a σ -field $\mathcal{F}$: (A1) $P(A) \geq 0$; (A2) $P(\Omega) = 1$; (A3) **countable** additivity: $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ for pairwise disjoint A_i .

1. **TRAP.** The axiom is *countable* additivity, not finite, and not “additivity over arbitrary (uncountable) unions”. Uncountable additivity is **false**: for $X \sim U(0,1)$, $1 = P(\Omega) \neq \sum_{x \in [0,1]} P(\{x\}) = 0$.
2. Countable additivity $\iff$ finite additivity + continuity from above at $\emptyset$ (if $A_n \downarrow \emptyset$ then $P(A_n) \rightarrow 0$). **MCQ angle:** “Which additional condition upgrades finite to countable additivity?”

3. Consequences you must be able to *derive*, since MCQs test the direction of each inequality: $P(\emptyset) = 0$; $P(A^c) = 1 - P(A)$; monotonicity $A \subseteq B \Rightarrow P(A) \leq P(B)$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; inclusion–exclusion for n events; Boole: $P(\cup A_i) \leq \sum P(A_i)$; Bonferroni: $P(A \cap B) \geq P(A) + P(B) - 1$, and generally $P(\cap_{i=1}^n A_i) \geq \sum P(A_i) - (n - 1)$.
4. Continuity of probability: $A_n \uparrow A \Rightarrow P(A_n) \uparrow P(A)$ and $A_n \downarrow A \Rightarrow P(A_n) \downarrow P(A)$. **TRAP.** Continuity from above needs no extra condition here because P is finite; in general measure theory it needs $\mu(A_1) < \infty$ — a favourite “statistics vs. measure” trick statement.
5. **TRAP.** $P(A) = 0$ does **not** imply $A = \emptyset$ (a continuous variable hits any fixed point with probability 0 yet hits *some* point). $P(A) = 1$ does not imply $A = \Omega$. Language: “almost surely”.
6. **TRAP.** $P(A) = P(B)$ does not imply $A = B$; $P(A \cap B) = 0$ does not imply $A \cap B = \emptyset$.
7. The domain matters: P is defined on a σ -field (closed under complements and *countable* unions), not on all subsets of Ω in general. **MCQ angle:** “The smallest σ -field containing A ” is $\{\emptyset, A, A^c, \Omega\}$ (4 sets); the smallest σ -field containing two “generic” events A, B has 16 sets; a σ -field over a finite Ω always has 2^k members for some k — never 3, 5, or 6 events. A quick kill: “Can a σ -field contain exactly 6 sets?” No.

2.3 Independence: the single richest MCQ mine in foundations

1. Definition: $P(A \cap B) = P(A)P(B)$. It is a property of the *measure*, not of the sets.
2. **TRAP. Mutually exclusive $\neq$ independent.** Disjoint events with $P(A) > 0, P(B) > 0$ are *never* independent ($0 \neq P(A)P(B) > 0$). They are, in fact, maximally dependent: knowing A happened kills B .
3. Any event with probability 0 or 1 is independent of *every* event, including itself. A independent of itself $\iff P(A) \in \{0, 1\}$ (since $P(A) = P(A)^2$).
4. If $A \perp B$ then automatically $A \perp B^c, A^c \perp B, A^c \perp B^c$.
5. **TRAP. Pairwise independence $\nRightarrow$ mutual independence.** Bernstein’s example: toss two fair coins; $A = \{\text{first is head}\}, B = \{\text{second is head}\}, C = \{\text{exactly one head}\}$. Each pair is independent ($P = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$), but $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.
6. **TRAP.** Conversely, $P(A \cap B \cap C) = P(A)P(B)P(C)$ alone does *not* give pairwise independence. Mutual independence of n events requires the product rule for **every** sub-collection: $2^n - n - 1$ equations.
7. Independence is *not* transitive: $A \perp B, B \perp C$ does not give $A \perp C$.
8. **TRAP.** Conditional independence neither implies nor is implied by independence. Classic: two coins, one fair and one two-headed, pick one at random; successive tosses are conditionally independent given the coin but unconditionally dependent.

3 Conditional Probability, Total Probability, Bayes — The Paradox Belt

3.1 Machinery with trap annotations

1. $P(A | B) = \frac{P(A \cap B)}{P(B)}$, defined **only when** $P(B) > 0$. **MCQ angle:** Statements quietly conditioning on probability-zero events are automatically suspect (see Borel–Kolmogorov below).

2. **KEY.** For fixed B , the map $A \mapsto P(A \mid B)$ is itself a full probability measure — so *every* identity (complement rule, inclusion–exclusion, Bayes) holds with an extra “ $\mid B$ ” everywhere. Many MCQs are ordinary identities wearing a conditional costume.
3. **TRAP.** $P(A \mid B) + P(A \mid B^c)$ need **not** equal 1. What equals 1 is $P(A \mid B) + P(A^c \mid B)$.
4. **TRAP.** No general order between $P(A \mid B)$ and $P(A)$: conditioning can raise, lower, or preserve probability. “ B occurred, so A is now more likely” is not a theorem.
5. Multiplication rule: $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cdots A_{n-1})$ — the backbone of urn/without-replacement questions.
6. Law of total probability: for a partition $\{B_i\}$ with $P(B_i) > 0$: $P(A) = \sum_i P(A \mid B_i)P(B_i)$. **TRAP.** The B_i must be exhaustive *and* mutually exclusive; examiners drop one of the two conditions in “which assumptions are needed” items.
7. Bayes: $P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}$. **TRAP.** Confusing $P(A \mid B)$ with $P(B \mid A)$ is the *prosecutor’s fallacy* — tested through medical-test and forensic wordings.
8. **KEY.** Symmetry principle for sampling without replacement: the unconditional probability that the k -th draw is red equals the proportion of reds initially, *for every* k . (Pólya urn versions also appear.) This one line solves a whole family of “second ball” questions in 10 seconds.

3.2 The paradox arsenal (each is a ready-made 2027 MCQ)

1. **Base-rate / false-positive paradox.** Disease prevalence 0.1%; test sensitivity 99%, specificity 99%. $P(\text{disease} \mid +) = \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.01 \times 0.999} \approx 9\%$, *not* 99%. **KEY.** When the prior is tiny, even excellent tests give mostly false positives.
2. **Monty Hall.** Three doors, host *knowingly* opens a goat door; switching wins with probability $\frac{2}{3}$. **TRAP.** If the host opens a door at random and it merely *happens* to show a goat, switching wins with probability $\frac{1}{2}$. The host’s knowledge is the entire problem. (100-door version: switching wins $\frac{99}{100}$.)
3. **Boy–girl paradox (order of information).** Two children, equally likely sexes. Given “at least one boy”: $P(\text{both boys}) = \frac{1}{3}$. Given “the elder is a boy”: $\frac{1}{2}$. Given “at least one boy born on a Tuesday”: $\frac{13}{27}$. **KEY.** *How* you learned the information changes the conditioning event.
4. **Bertrand’s box paradox.** Three boxes: GG, SS, GS. A random coin from a random box is gold; $P(\text{other coin gold}) = \frac{2}{3}$, *not* $\frac{1}{2}$ (count *coins*, not boxes).
5. **Bertrand’s chord paradox.** “A random chord of a circle is longer than the side of the inscribed equilateral triangle” has answers $\frac{1}{3}$ (random endpoints), $\frac{1}{2}$ (random midpoint on a fixed radius), $\frac{1}{4}$ (random midpoint in the disc). **KEY.** Moral: “at random” in continuous settings is meaningless until the randomising mechanism is fixed — the standard MCQ asks you to match method to answer.
6. **Simpson’s paradox.** Treatment A can beat B inside *every* subgroup yet lose on aggregate, because group sizes act as hidden weights. **MCQ angle:** It is possible that $P(S \mid A \cap C) > P(S \mid B \cap C)$ and $P(S \mid A \cap C^c) > P(S \mid B \cap C^c)$, yet $P(S \mid A) < P(S \mid B)$ — **True** (the two treatments meet the subgroups with different weights). **TRAP.** But with a *common* partition and common weights, $P(A \mid C) > P(B \mid C)$ and $P(A \mid C^c) > P(B \mid C^c)$ *do* force $P(A) > P(B)$ — the exam plays these two lookalike statements against each other.

7. **Gambler’s fallacy.** Independent tosses have no memory: after 10 heads, $P(\text{head})$ is still $\frac{1}{2}$. Contrast deliberately with *memorylessness* of geometric/exponential (which is a property of those laws, not of “luck”).
8. **Two-envelope paradox.** The bogus computation $E(\text{other}) = \frac{1}{2}(2x) + \frac{1}{2}(x/2) = 1.25x$ illegally treats x as fixed under both hypotheses / assumes an improper uniform prior on all amounts. **MCQ angle:** “Where does the argument break?”
9. **Borel–Kolmogorov paradox.** Conditional distributions given a probability-zero event (a point on a great circle of a uniformly random point on a sphere) are *not intrinsically defined*: different limiting parametrisations give different answers. **KEY.** Conditioning must be on a σ -field/partition, not on an isolated null event.
10. **de Méré’s paradox.** At least one six in 4 throws: $1 - (5/6)^4 \approx 0.518 > \frac{1}{2}$; at least one double-six in 24 double throws: $1 - (35/36)^{24} \approx 0.491 < \frac{1}{2}$. Naive proportionality (4/6 vs 24/36) fails.
11. **Birthday paradox.** 23 people suffice for $P(\text{shared birthday}) > \frac{1}{2}$ (≈ 0.507). **TRAP.** The related but different question “someone shares *my* birthday” needs 253 people — examiners swap the two.
12. **St. Petersburg paradox.** Payoff 2^n with probability 2^{-n} : expectation $= \infty$ although the game is “obviously” worth little; shows the mean may fail to exist/be infinite and cannot always guide decisions. (Full treatment with expectations in §8.)

4 Random Variables and Distribution Functions — Fine Print That Becomes MCQs

4.1 Distribution function (CDF): the exact properties

$F(x) = P(X \leq x)$ satisfies, and is characterised by: (i) non-decreasing; (ii) **right**-continuous; (iii) $F(-\infty) = 0$, $F(+\infty) = 1$. Conversely, *every* function with (i)–(iii) is the CDF of some random variable (correspondence theorem).

1. **TRAP.** Right-continuous, *not* necessarily left-continuous. The jump at x is exactly the point mass: $P(X = x) = F(x) - F(x^-)$. Also $P(X < x) = F(x^-)$, $P(a < X \leq b) = F(b) - F(a)$, $P(a \leq X \leq b) = F(b) - F(a^-)$. Examiners live inside these strict-vs-weak inequality distinctions.
2. A CDF has at most **countably many** discontinuities (jumps of size $\geq 1/n$ number at most n). **MCQ angle:** “Can a CDF have uncountably many jumps?” No.
3. **TRAP.** “ F continuous $\Rightarrow$ density exists” is **FALSE**. The Cantor (Lebesgue singular) distribution has a continuous CDF, no point masses, yet no density: $F' = 0$ almost everywhere while F climbs from 0 to 1. Lebesgue decomposition: every CDF = mixture of discrete + absolutely continuous + singular continuous parts. “Continuous random variable” in exam usage means *absolutely* continuous (density exists) — but a top-1% item can hinge on the distinction.
4. **TRAP.** A pdf may exceed 1 (e.g. $U(0, \frac{1}{2})$ has $f = 2$; $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ is even unbounded). Only $f \geq 0$ and $\int f = 1$ are required. Contrast: a pmf can never exceed 1.
5. A pdf is not unique — altering it on a set of measure zero changes nothing. And f need not be continuous or bounded.

6. Mixed distributions exist and are examinable: e.g. F with a jump at 0 and a density on $(0, \infty)$ (rainfall; waiting time with $P(X=0) > 0$). For these, neither pure $\sum$ nor pure $\int$ computes moments alone.
7. For continuous F : $P(X = x) = 0$ for every x — yet X always takes *some* value. “Probability zero” $\neq$ “impossible”.
8. Symmetry about 0 (continuous case): $F(-x) = 1 - F(x)$; median = 0; all existing odd moments vanish; the characteristic function is real (§9).
9. Quantiles: $Q(p) = \inf\{x : F(x) \geq p\}$. **TRAP.** The median need not be unique (any point of a flat stretch of F at level $\frac{1}{2}$); conventionally the midpoint is quoted.

4.2 Building new CDFs from old — a favourite “which of these is a CDF” battery

Let F, G be CDFs. Then:

Candidate	CDF?	Reason / interpretation
$\alpha F + (1 - \alpha)G, 0 \leq \alpha \leq 1$	Yes	mixture distribution
FG	Yes	CDF of $\max(X, Y), X \perp Y$
$F^n \ (n \in \mathbb{N})$	Yes	max of n iid copies
$1 - (1 - F)^n$	Yes	min of n iid copies
$F + G$	No	tends to 2
$F(ax + b), a > 0$	Yes	location–scale change
$F(ax + b), a < 0$	No	decreasing
$F(G(x))$	TRAP. No in general	limits are $F(0) \geq 0$ and $F(1) \leq 1$, not 0, 1; valid iff F puts all mass on $[0, 1]$ (e.g. F uniform/Beta)
$1 - F(-x^-)$	Yes	CDF of $-X$

KEY. Test the three axioms mechanically; interpret when possible (max/min interpretations also produce direct questions on order statistics).

4.3 Random variables: definitional traps

1. A random variable is a (measurable) *function* $\Omega \rightarrow \mathbb{R}$; it is neither random nor a variable — assertion–reason bait.
2. A constant is a legitimate random variable (degenerate distribution; CDF is a unit step). Its variance is 0, and $\text{Var}(X) = 0 \iff X$ is a.s. constant.
3. **TRAP.** Two different random variables can have the *same* distribution without being equal: $X \sim N(0, 1)$ and $-X$ have identical laws but $P(X = -X) = P(X = 0) = 0$. “Equal in distribution” $\neq$ “equal”. MCQs phrase this as “ $X \stackrel{d}{=} Y \Rightarrow X = Y$?” — False.
4. If X has CDF F *continuous*, then $F(X) \sim U(0, 1)$ (probability integral transform); conversely $F^{-1}(U)$ has CDF F (inverse transform / simulation). **TRAP.** Continuity of F is essential for the first direction; the second works for any F via the generalised inverse.
5. Existence of moments is a property of the *tail* of F . Heavier tail $\Rightarrow$ fewer moments. The Cauchy has none (not even the mean); t -type tails with index α have moments of order $< \alpha$ only.

5 Standard Distributions — Distribution-by-Distribution Ambush Points

KEY. For every family memorise five hooks: (support, mean, variance, generating function, one *characterising* property). The characterising property is what the top-1% items test.

5.1 Bernoulli(p)

1. $\mathbb{E}X = p$, $\text{Var}X = pq \leq \frac{1}{4}$, maximised at $p = \frac{1}{2}$. **MCQ angle:** “Largest possible variance of a 0–1 variable” = $\frac{1}{4}$.
2. $X^k = X$ for all $k \geq 1$: every raw moment equals p . A one-line MCQ.
3. Indicator calculus: $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$, $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B$; expectation of an indicator is a probability — the trick behind “matching/coupon” expectation items.

5.2 Binomial(n, p)

1. MGF $(q + pe^t)^n$; PGF $(q + ps)^n$. Mean np **always exceeds** variance npq (underdispersion). **MCQ angle:** “mean = 8, variance = 16” is impossible for a binomial — instant elimination.
2. If mean $np = a$ and variance $npq = b$ then $q = b/a$, $p = 1 - b/a$, $n = a^2/(a - b)$; n must come out a positive integer, else no such binomial exists.
3. Mode: $\lfloor (n+1)p \rfloor$; if $(n+1)p$ is an integer, *two* modes: $(n+1)p$ and $(n+1)p - 1$. Recurrence $\frac{P(X=k+1)}{P(X=k)} = \frac{n-k}{k+1} \cdot \frac{p}{q}$.
4. Additivity: $\text{Bin}(m, p) + \text{Bin}(n, p) = \text{Bin}(m+n, p)$ for independent summands **TRAP.** *only with common p* . Different p ’s give a non-binomial law.
5. Symmetric iff $p = \frac{1}{2}$. $n - X \sim \text{Bin}(n, q)$.
6. Limits: $n \rightarrow \infty$, $np \rightarrow \lambda$: Poisson (law of rare events); fixed p , large n : normal (de Moivre–Laplace) with continuity correction $P(X \leq k) \approx \Phi\left(\frac{k+0.5-np}{\sqrt{npq}}\right)$. **MCQ angle:** Poisson approximation is good when n large, p small; normal when npq large.

5.3 Poisson(λ)

1. Mean = variance = λ : the *equidispersion* signature. Also $\mathbb{E}X = \mathbb{E}X(X-1) + \dots$ trick: factorial moments $\mathbb{E}X(X-1)\cdots(X-r+1) = \lambda^r$.
2. Mode $\lfloor \lambda \rfloor$; two modes $\lambda - 1, \lambda$ when λ is a positive integer. Recurrence $P(X = k+1) = \frac{\lambda}{k+1} P(X = k)$.
3. PGF $e^{\lambda(s-1)}$; MGF $e^{\lambda(e^t-1)}$. Additive: independent Poissons sum to Poisson (rates add). **KEY.** Raikov’s theorem (converse, top-1%): if $X \perp Y$, both non-degenerate, and $X + Y$ is Poisson, then X and Y are each Poisson.
4. **TRAP.** The *difference* of independent Poissons is **not** Poisson (it can be negative; it is Skellam). Constant multiples cX ($c \neq 1$) are not Poisson either — “ $2X$ is Poisson with mean 2λ ” is False.
5. Conditional: $X \mid X + Y = n \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$ — a recurring exam favourite.
6. Thinning: independently keeping each event with probability p turns Poisson(λ) into Poisson($p\lambda$); the kept and discarded streams are *independent* (surprising — classic top-end MCQ).

7. $P(X \text{ even}) = e^{-\lambda} \cosh \lambda = \frac{1}{2}(1 + e^{-2\lambda}) > \frac{1}{2}$.
8. Poisson $\rightarrow$ Normal as $\lambda \rightarrow \infty$ (standardised); continuity correction applies.

5.4 Geometric(p) — the discrete memoryless law

1. **TRAP.** Two conventions. Trials to first success, support $\{1, 2, \dots\}$: $P(X = k) = q^{k-1}p$, $\mathbb{E}X = 1/p$, $\text{Var}X = q/p^2$, $P(X > k) = q^k$. Failures before first success, support $\{0, 1, \dots\}$: $\mathbb{E}Y = q/p$, same variance. Check the support before answering *any* numeric item.
2. Memorylessness $P(X > m + n \mid X > m) = P(X > n)$ characterises the geometric among discrete laws on $\{1, 2, \dots\}$ (and the exponential among continuous laws). “The only memoryless discrete distribution” — True (with that support).
3. $\min(X_1, X_2)$ of independent geometrics is geometric with $q = q_1 q_2$.
4. Mode is always the smallest support point (1 or 0); the pmf is strictly decreasing.

5.5 Uniform / Rectangular

1. **TRAP.** Two objects share the name. **Discrete** uniform on $\{1, \dots, n\}$: mean $\frac{n+1}{2}$, variance $\frac{n^2-1}{12}$. **Continuous** (rectangular) $U(a, b)$: mean $\frac{a+b}{2}$, variance $\frac{(b-a)^2}{12}$. The two variance formulas are deliberately confusable.
2. $U(a, b)$: MGF exists everywhere: $\frac{e^{bt}-e^{at}}{(b-a)t}$ ($t \neq 0$); all moments finite. No unique mode.
3. Sum of two iid $U(0, 1)$ is **triangular** on $(0, 2)$ — not uniform. $X + Y$ uniform is impossible for iid X, Y (no distribution convolves with itself to a uniform).
4. $-\ln U \sim \text{Exp}(1)$; $F^{-1}(U)$ simulates F ; $F(X) \sim U(0, 1)$ for continuous F (probability integral transform).
5. Order statistics: $U_{(k)} \sim \text{Beta}(k, n+1-k)$; $\mathbb{E}U_{(k)} = \frac{k}{n+1}$; range $R = U_{(n)} - U_{(1)} \sim \text{Beta}(n-1, 2)$.
6. Maximum entropy on a bounded interval.

5.6 Exponential(λ) — the continuous memoryless law

1. Mean $1/\lambda$, variance $1/\lambda^2$, median $\frac{\ln 2}{\lambda} \approx 0.693/\lambda < \text{mean}$; $P(X > \mathbb{E}X) = e^{-1} \approx 0.368$ (not $\frac{1}{2}$ — a beloved trap). Mode = 0.
2. Memorylessness characterises it among continuous laws on $(0, \infty)$; equivalently constant hazard rate λ . “Used/unused component equally good” wordings.
3. $\min(X_1, \dots, X_n) \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$; $P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$; the minimum and the index achieving it are *independent* (top-1% fact).
4. **TRAP.** Sums are *not* exponential: $\sum_{i=1}^n \text{iid Exp}(\lambda) = \text{Gamma/Erlang}(n, \lambda)$. The *max* of iid exponentials is not exponential either.
5. $X_1 - X_2$ (iid) $\sim \text{Laplace}$; $2\lambda X \sim \chi_2^2$; interarrival times of a Poisson process are iid exponential — the bridge fact.
6. Maximum entropy on $(0, \infty)$ given the mean.

5.7 Normal(μ, σ^2)

- MGF $e^{\mu t + \sigma^2 t^2/2}$ (exists everywhere); mean = median = mode; inflection points at $\mu \pm \sigma$; central moments: odd all 0, $\mathbb{E}(X - \mu)^{2k} = \sigma^{2k} (2k - 1)!!$, so $\mathbb{E}(X - \mu)^4 = 3\sigma^4$ (kurtosis 3, excess 0).
- Coverage: $\mu \pm \sigma$: 68.27%, $\mu \pm 2\sigma$: 95.45%, $\mu \pm 3\sigma$: 99.73%. Mean deviation = $\sigma \sqrt{2/\pi} \approx 0.7979\sigma$; QD $\approx 0.6745\sigma$; QD : MD : SD $\approx 10:12:15$.
- Any linear combination of *independent* normals is normal: $aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$. **TRAP**. Variances add with a^2, b^2 — for $X - Y$ the variance is $\sigma_1^2 + \sigma_2^2$ (never subtract).
- TRAP**. X normal, Y normal $\nRightarrow X + Y$ normal, unless independent or *jointly* normal. Counterexample: $Y = X \cdot S$ with $S = \pm 1$ a fair sign flip independent of $X \sim N(0, 1)$: $Y \sim N(0, 1)$, but $X + Y$ has an atom at 0 ($P(X + Y = 0) = \frac{1}{2}$), so it is not normal. Same example shows: marginally normal, uncorrelated, yet *dependent* — “uncorrelated normals are independent” is **False** without joint normality.
- Characterisations (top-1% material): **Cramér**: if $X \perp Y$ and $X + Y$ is normal, each is normal. **Bernstein**: if X, Y iid and $X + Y \perp X - Y$, then they are normal. For iid $N(\mu, \sigma^2)$, $X + Y$ and $X - Y$ are independent — special to the normal. Maximum entropy for fixed variance. $\bar{X} \perp S^2$ iff normal.
- $Z^2 \sim \chi_1^2 = \text{Gamma}(\frac{1}{2}, \frac{1}{2})$; $X/Y \sim \text{Cauchy}$ for iid $N(0, 1)$; $|Z|$ is half-normal with $\mathbb{E}|Z| = \sqrt{2/\pi}$; $\bar{X}_n \sim N(\mu, \sigma^2/n)$ *exactly*, at every n .
- $\Phi(-x) = 1 - \Phi(x)$; standardisation $Z = (X - \mu)/\sigma$ — sign errors here are the cheapest marks lost in the exam.

5.8 Cauchy(θ, σ) — the professional counterexample

- Density $\frac{1}{\pi} \cdot \frac{\sigma}{\sigma^2 + (x - \theta)^2}$. **No mean** (the defining integral diverges: undefined, not “0 by symmetry” — a razor-sharp trap), hence no variance, no higher moments, **no MGF at any $t \neq 0$** . Median = mode = θ ; quartiles $\theta \pm \sigma$; use quartile deviation, not SD.
- CF: $\varphi(t) = e^{i\theta t - \sigma|t|}$ — exists (CFs always do), but is not differentiable at 0, matching the nonexistent mean.
- $\bar{X}_n \sim \text{Cauchy}(\theta, \sigma)$ for every n : averaging does **nothing**. Hence WLLN, SLLN and CLT all fail; the sample median, not the sample mean, estimates θ .
- Stable family: independent Cauchys add to a Cauchy (location and scale both add). $1/X$, $\tan(U)$ for $U \sim U(-\frac{\pi}{2}, \frac{\pi}{2})$, and the ratio of iid $N(0, 1)$ are standard Cauchy; Cauchy = t -distribution with 1 degree of freedom.

5.9 Hypergeometric(N, K, n)

- Sampling *without* replacement: draws are dependent. Mean $n \frac{K}{N}$ (same as binomial); variance $n \frac{K}{N} (1 - \frac{K}{N}) \frac{N-n}{N-1}$ — binomial variance shrunk by the finite population correction $\frac{N-n}{N-1} < 1$. **MCQ angle**: “Same mean, smaller variance than the matching binomial” — True.
- As $N \rightarrow \infty$ with $K/N \rightarrow p$: converges to $\text{Bin}(n, p)$ — justifies treating draws from huge populations as independent.
- TRAP**. By symmetry the *unconditional* law of each single draw is Bernoulli(K/N) even without replacement — independence fails, identical distribution does not.

5.10 Multinomial($n; p_1, \dots, p_k$)

1. Marginals: $X_i \sim \text{Bin}(n, p_i)$. **TRAP.** Components are *negatively* correlated by the constraint $\sum X_i = n$: $\text{Cov}(X_i, X_j) = -np_i p_j$, $\rho_{ij} = -\sqrt{\frac{p_i p_j}{(1-p_i)(1-p_j)}}$ — the sign question is a standard MCQ.
2. Conditionals stay in the family: $X_i \mid X_j = m \sim \text{Bin}(n - m, \frac{p_i}{1-p_j})$; lumping cells keeps it multinomial.
3. The joint pmf coefficient is $\binom{n}{x_1 \dots x_k} = \frac{n!}{x_1! \dots x_k!}$; for $k = 2$ it collapses to the binomial — “binomial is a multinomial with $k = 2$ ”.

5.11 Laplace (double exponential)(μ, b)

1. Density $\frac{1}{2b}e^{-|x-\mu|/b}$: a symmetric *peak* (not smooth) at μ . Mean = median = mode = μ ; variance $2b^2$ (**TRAP.** not b^2); kurtosis 6 (excess 3): sharply leptokurtic — “more peaked *and* heavier-tailed than normal”.
2. $X_1 - X_2$ for iid exponentials is Laplace — the constructive definition MCQs use.
3. CF $\frac{e^{i\mu t}}{1+b^2 t^2}$; MGF exists only for $|t| < 1/b$.
4. MLE of μ is the sample **median** (contrast: normal $\rightarrow$ mean, uniform $\rightarrow$ midrange) — a three-way matching question waiting to happen.

5.12 Negative Binomial(r, p)

1. **TRAP.** Two conventions again. Trials until r -th success X (support $\{r, r+1, \dots\}$): $\mathbb{E}X = r/p$, $\text{Var} = rq/p^2$. Failures before r -th success Y (support $\{0, 1, \dots\}$): $\mathbb{E}Y = rq/p$, $\text{Var} = rq/p^2$. Variance $>$ mean (**overdispersion**) — the mirror image of the binomial; the “mean vs variance” triage (binomial $<$, Poisson $=$, negative binomial $>$) is a guaranteed MCQ.
2. Sum of r iid geometrics; closed under independent addition with common p (add the r ’s).
3. Gamma-mixed Poisson is negative binomial — the standard model for overdispersed counts (top-1% link fact).
4. Limit: $r \rightarrow \infty, q \rightarrow 0, rq \rightarrow \lambda$ gives Poisson(λ).

5.13 Beta(α, β)

1. Support $(0, 1)$; mean $\frac{\alpha}{\alpha+\beta}$; variance $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$; mode $\frac{\alpha-1}{\alpha+\beta-2}$ for $\alpha, \beta > 1$.
2. Shape triage: $\alpha = \beta$ symmetric; both < 1 : U-shaped (arcsine Beta($\frac{1}{2}, \frac{1}{2}$) has *unbounded* density — densities need not be bounded); Beta(1, 1) = $U(0, 1)$.
3. $1 - X \sim \text{Beta}(\beta, \alpha)$; uniform order statistics $U_{(k)} \sim \text{Beta}(k, n+1-k)$.
4. Gamma bridge: for independent Gamma(α_1, λ), Gamma(α_2, λ): $\frac{X}{X+Y} \sim \text{Beta}(\alpha_1, \alpha_2)$ and is **independent of** $X + Y$ (Lukacs’ characterisation of the gamma — elite MCQ material).

5.14 $\text{Gamma}(\alpha, \lambda)$ (shape–rate)

1. Mean α/λ , variance α/λ^2 ; MGF $(1 - t/\lambda)^{-\alpha}$ **TRAP**. valid only for $t < \lambda$ — “MGF exists for all t ” is False. Skewness $2/\sqrt{\alpha} \rightarrow 0$: gamma $\rightarrow$ normal as $\alpha \rightarrow \infty$.
2. **TRAP**. Shape–rate vs shape–scale parametrisations differ by $\lambda \leftrightarrow 1/\theta$; check which one the question uses before computing anything.
3. Additivity in the *shape* parameter for common rate: $\text{Gamma}(\alpha_1, \lambda) + \text{Gamma}(\alpha_2, \lambda) = \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$. Special cases: exponential = $\text{Gamma}(1, \lambda)$; $\chi_k^2 = \text{Gamma}(k/2, \frac{1}{2})$.
4. Scaling: $cX \sim \text{Gamma}(\alpha, \lambda/c)$ — shape is scale-invariant.
5. Mode $\frac{\alpha-1}{\lambda}$ for $\alpha \geq 1$; for $\alpha < 1$ the density blows up at 0.

5.15 $\text{Lognormal}(\mu, \sigma^2)$

1. $X = e^Z$, $Z \sim N(\mu, \sigma^2)$. **TRAP**. μ, σ^2 are the mean/variance of $\ln X$, *not* of X . Median e^μ ; mean $e^{\mu + \sigma^2/2}$; mode $e^{\mu - \sigma^2}$; always mode < median < mean (right skew).
2. All moments finite: $\mathbb{E}X^k = e^{k\mu + k^2\sigma^2/2}$ — **yet the MGF does not exist for any $t > 0$** . “All moments exist $\Rightarrow$ MGF exists” is **False**; the lognormal is the canonical counterexample.
3. **TRAP**. The lognormal is **not determined by its moments**: distinct distributions share all its moments (moment problem indeterminacy). “Equal moments of all orders $\Rightarrow$ equal distributions” is **False** in general (true if an MGF exists in a neighbourhood of 0).
4. Products (not sums) of independent lognormals are lognormal; multiplicative CLT: products of many positive iid factors are asymptotically lognormal.

5.16 Cross-family cheat tables (matching-type questions)

Closed under independent summation: Binomial (common p), Poisson, Normal, Gamma (common rate; includes χ^2), Cauchy, Negative binomial (common p).

Not closed: Uniform (sum triangular), Exponential (sum Gamma), Geometric (sum NB), Bernoulli (sum Binomial), Beta, Laplace, Hypergeometric, Lognormal (products instead).

Limit lattice: Hypergeometric $\rightarrow$ Binomial ($N \rightarrow \infty$); Binomial $\rightarrow$ Poisson ($n \rightarrow \infty, np \rightarrow \lambda$); Binomial $\rightarrow$ Normal (de Moivre–Laplace); Poisson $\rightarrow$ Normal ($\lambda \rightarrow \infty$); Negative binomial $\rightarrow$ Poisson ($r \rightarrow \infty, rq \rightarrow \lambda$); Gamma $\rightarrow$ Normal ($\alpha \rightarrow \infty$); Geometric (rescaled) $\rightarrow$ Exponential; $t_\nu \rightarrow$ Normal ($\nu \rightarrow \infty$) with $t_1 = \text{Cauchy}$.

Mean vs variance triage (count data): Binomial: Var < Mean; Poisson: Var = Mean; Negative binomial: Var > Mean.

Memorylessness: Geometric (only discrete), Exponential (only continuous). No other distribution qualifies — any MCQ claiming otherwise is False.

Maximum entropy: bounded support $\rightarrow$ Uniform; positive with fixed mean $\rightarrow$ Exponential; fixed mean and variance on $\mathbb{R} \rightarrow$ Normal.

MGF existence: everywhere: Normal, Binomial, Poisson, Uniform, Bernoulli; in a neighbourhood only: Exponential/Gamma ($t < \lambda$), Laplace ($|t| < 1/b$), Geometric/NB ($e^t < 1/q$); nowhere useful: Cauchy (no $t \neq 0$), Lognormal (no $t > 0$). CFs exist for *all* of them, always.

6 Random Vectors: Joint, Marginal, Conditional — and Transformations

6.1 Joint vs marginal: the one-way street

1. The joint distribution determines all marginals; **TRAP**. marginals **never** determine the joint (except under independence). Canonical counterexample: both (X, Y) bivariate normal with $\rho = 0$, and the “sign-flipped” pair (X, SX) , have standard normal marginals with entirely different joint laws. “Standard normal marginals $\Rightarrow$ bivariate normal” is **False**.
2. Joint CDF $F(x, y)$ properties: non-decreasing and right-continuous in each argument, $F(-\infty, y) = F(x, -\infty) = 0$, $F(\infty, \infty) = 1$, *plus* the rectangle inequality $F(b, d) - F(a, d) - F(b, c) + F(a, c) \geq 0$. **TRAP**. The rectangle condition is the one candidates forget — monotone-in-each-argument alone is *not* sufficient to be a joint CDF (e.g. $F(x, y) = \min(1, \max(0, x + y))$ -type examples fail it).
3. Marginal extraction: $F_X(x) = F(x, \infty)$; $f_X(x) = \int f(x, y) dy$. Conditional density $f_{Y|X}(y | x) = f(x, y)/f_X(x)$ for $f_X(x) > 0$ — it is a genuine density in y (integrates to 1) — and *conditioning on* $\{X = x\}$, a null event, is defined through this density, not through the elementary formula (Borel–Kolmogorov again).

6.2 Independence of random variables: precision points

1. Definition: $F(x, y) = F_X(x)F_Y(y)$ for **all** (x, y) ; equivalently the density/pmf factorises for (almost) all points.
2. **KEY**. Factorisation shortcut: if $f(x, y) = g(x)h(y)$ on a **rectangular** support (product set), then $X \perp Y$ (no need to normalise g, h). **TRAP**. If the support is *not* a product set — e.g. $f(x, y) = 2$ on the triangle $0 < x < y < 1$, or uniform on a disc — the variables are dependent *no matter what* the formula looks like. Support check first, always.
3. Uniform on the disc $x^2 + y^2 \leq 1$: X, Y uncorrelated but dependent (the range of Y depends on X). Uniform on the *square*: independent. A ready-made pair of MCQ options.
4. $X \perp Y \Rightarrow g(X) \perp h(Y)$ for any (measurable) g, h . **TRAP**. The converse direction of specific instances fails; and $X \perp Y, X \perp Z$ do **not** give $X \perp (Y, Z)$ or $X \perp Y + Z$ in general.
5. Uncorrelated $\neq$ independent: $X \sim N(0, 1), Y = X^2$: $\text{Cov}(X, Y) = \mathbb{E}X^3 = 0$, yet Y is a function of X . Independence $\Rightarrow$ uncorrelated (when moments exist); the converse holds inside a *jointly* normal vector — that “jointly” is the entire MCQ.

6.3 Distributions of functions of random variables

1. Monotone change of variable: $Y = g(X)$, g strictly monotone with inverse g^{-1} : $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$. **TRAP**. Forgetting the absolute value, or using dg/dx instead of dg^{-1}/dy , are the two standard planted wrong options.
2. Non-monotone: $Y = X^2$: $f_Y(y) = \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})]$, $y > 0$. With $X \sim N(0, 1)$ this gives χ_1^2 — derive it once, remember it forever.
3. Bivariate (Jacobian) method: $(U, V) = T(X, Y)$ invertible: $f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v)) |J|$ where $J = \det \frac{\partial(x, y)}{\partial(u, v)}$. **TRAP**. The Jacobian of the *inverse* map appears; equivalently $|J| = 1/|J_{\text{forward}}|$.

4. Convolution: $X \perp Y \Rightarrow f_{X+Y}(z) = \int f_X(x)f_Y(z-x)dx$. **TRAP.** Get limits from the supports; for positive variables the integral runs over $0 < x < z$ only.
5. Standard function-of-variable identities (memorise as a table): $Z^2 \sim \chi_1^2$; $X/Y \sim \text{Cauchy}$ (iid $N(0,1)$); $-\frac{1}{\lambda} \ln U \sim \text{Exp}(\lambda)$; $F(X) \sim U(0,1)$; $\frac{X}{X+Y} \sim \text{Beta}$ (independent gammas, common rate), independent of $X+Y$; min of exponentials exponential; $X_1 - X_2$ of iid exponentials Laplace; Box-Muller: $\sqrt{-2 \ln U_1} \cos(2\pi U_2)$ and $\sqrt{-2 \ln U_1} \sin(2\pi U_2)$ are iid $N(0,1)$.
6. Order statistics (iid, CDF F , density f): $F_{X_{(n)}} = F^n$, $F_{X_{(1)}} = 1 - (1 - F)^n$; $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F^{k-1}(1 - F)^{n-k} f(x)$. **TRAP.** max and min of iid variables are *not* independent, and the identity $X_{(1)} + X_{(n)} \stackrel{d}{=} X_1 + X_n$ is false in general.
7. $\mathbb{E}g(X)$ via the law of the unconscious statistician: no need to find the law of $g(X)$ first — but **TRAP.** $\mathbb{E}g(X) \neq g(\mathbb{E}X)$ in general (Jensen direction: $\geq$ for convex g , $\leq$ for concave).

7 Modes of Convergence — The Counterexample Arsenal

7.1 The hierarchy (memorise as a picture)

$$\boxed{X_n \xrightarrow{\text{a.s.}} X} \Rightarrow \boxed{X_n \xrightarrow{P} X} \Rightarrow \boxed{X_n \xrightarrow{d} X} \quad \text{and} \quad \boxed{X_n \xrightarrow{L^2} X} \Rightarrow \boxed{X_n \xrightarrow{P} X}$$

No other implication holds in general. **KEY.** Two special upgrades: (i) $X_n \xrightarrow{d} c$ (a **constant**) $\Rightarrow X_n \xrightarrow{P} c$; (ii) $X_n \xrightarrow{P} X \Rightarrow$ some *subsequence* converges a.s. Also, a.s. convergence and L^2 convergence do not imply each other.

7.2 Definitions with the exact quantifier traps

1. In probability: $\forall \varepsilon > 0, P(|X_n - X| > \varepsilon) \rightarrow 0$. The probability of a *single* large deviation dies.
2. Almost surely: $P(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1$. The entire *path* converges, except on a null set. Equivalent tail form: $P(\sup_{k \geq n} |X_k - X| > \varepsilon) \rightarrow 0$.
3. In mean square: $\mathbb{E}|X_n - X|^2 \rightarrow 0$ (needs second moments). L^r convergence $\Rightarrow L^s$ for $s < r$ (Lyapunov), and $\Rightarrow$ convergence of $\mathbb{E}|X_n|^r \rightarrow \mathbb{E}|X|^r$.
4. In distribution: $F_n(x) \rightarrow F(x)$ at every **continuity point** of F . **TRAP.** Convergence at discontinuity points is *not* required: $X_n = 1/n$ (degenerate) $\xrightarrow{d} 0$, yet $F_n(0) = 0 \not\rightarrow F(0) = 1$. This is a certified classic MCQ.
5. **TRAP.** Convergence in distribution is a statement about *laws*, not values: X_n and X need not even live on the same probability space, and $X_n \xrightarrow{d} X, Y_n \xrightarrow{d} Y$ does **not** imply $X_n + Y_n \xrightarrow{d} X + Y$ (unless e.g. independence throughout, or one limit is constant — Slutsky).

7.3 The four counterexamples that answer 90% of convergence MCQs

1. **Typewriter / sliding blocks** ($\xrightarrow{P}$ and $\xrightarrow{L^2}$ but *not* a.s.): on $[0,1]$ with Lebesgue measure, let X_n be indicators of intervals of shrinking length $1/k$ sweeping repeatedly across $[0,1]$. Then $P(X_n \neq 0) \rightarrow 0$ so $X_n \xrightarrow{P} 0$ and $\mathbb{E}X_n^2 \rightarrow 0$, but every ω is hit infinitely often: $X_n(\omega)$ does not converge *anywhere*.
2. **Tall spike** ($\xrightarrow{\text{a.s.}}$ and $\xrightarrow{P}$ but *not* in mean square): $X_n = n \mathbf{1}_{(0,1/n)}(U)$, $U \sim U(0,1)$. Then $X_n \rightarrow 0$ a.s. (for each ω , eventually $U > 1/n$), but $\mathbb{E}X_n = 1$ for all n and $\mathbb{E}X_n^2 = n \rightarrow \infty$:

no L^1 or L^2 convergence. **KEY.** Moral: convergence a.s. says nothing about moments — “ $X_n \xrightarrow{\text{a.s.}} X \Rightarrow \mathbb{E}X_n \rightarrow \mathbb{E}X$ ” is **False** (needs uniform integrability or dominated convergence).

- Independent rare events** ($\xrightarrow{P}$ but *not* a.s.): independent $X_n \in \{0, 1\}$ with $P(X_n = 1) = 1/n$. Then $X_n \xrightarrow{P} 0$; but $\sum 1/n = \infty$ and independence, so by Borel–Cantelli II $X_n = 1$ infinitely often a.s.: no a.s. limit. **TRAP.** With $P(X_n = 1) = 1/n^2$ instead, $\sum < \infty$: Borel–Cantelli I gives $X_n \xrightarrow{\text{a.s.}} 0$. The exponent 1 vs 2 is exactly where the MCQ lives.
- Symmetric flip** ($\xrightarrow{d}$ but *not* $\xrightarrow{P}$): $X \sim N(0, 1)$ and $X_n = -X$ for all n . Then $X_n \stackrel{d}{=} X$ so trivially $X_n \xrightarrow{d} X$, but $|X_n - X| = 2|X|$ never shrinks. Convergence of distributions is not convergence of random variables.

7.4 Theorems the examiner expects you to wield in one line

- Slutsky:** $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{P} c \Rightarrow X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, $X_n/Y_n \xrightarrow{d} X/c$ ($c \neq 0$). **TRAP.** Y_n must tend to a *constant*, not a random limit.
- Continuous mapping:** $X_n \xrightarrow{d} X \Rightarrow g(X_n) \xrightarrow{d} g(X)$ for g continuous (a.s. w.r.t. the limit law); same statement for $\xrightarrow{P}$ and $\xrightarrow{\text{a.s.}}$.
- Pólya:** if $F_n \rightarrow F$ pointwise and F is *continuous*, the convergence is automatically *uniform* in x — an elegant true statement often planted among false ones.
- Scheffé:** pointwise convergence of *densities* $\Rightarrow$ convergence in distribution (indeed in total variation). **TRAP.** The converse fails: $\xrightarrow{d}$ does not imply density convergence (densities may not even exist).
- Monotone relationships: $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y \Rightarrow X = Y$ a.s. (limits in probability are a.s. unique). Limits in distribution are unique only *as laws*.

8 Expectation and Conditional Expectation

8.1 Existence and computation traps

- $\mathbb{E}X$ exists (finitely) iff $\mathbb{E}|X| < \infty$. **TRAP.** For the Cauchy, $\int_{-a}^a x f(x) dx = 0$ for every a , yet the mean is *undefined* — symmetry is not permission. “Cauchy has mean 0 by symmetry” is the single most planted false option in this chapter.
- St. Petersburg: payoff 2^n w.p. 2^{-n} ($n = 1, 2, \dots$): $\mathbb{E}X = \sum 1 = \infty$. Infinite expectations are legal; the LLN conclusion then reads $\bar{X}_n \rightarrow \infty$ a.s.
- Existence ladder (Lyapunov): $\mathbb{E}|X|^r < \infty \Rightarrow \mathbb{E}|X|^s < \infty$ for all $0 < s < r$. Higher moments imply lower — **TRAP.** the reverse claim is False (t_3 has a mean and variance but no third moment).
- Tail formulas: $X \geq 0$: $\mathbb{E}X = \int_0^\infty P(X > x) dx$; integer-valued: $\mathbb{E}X = \sum_{k \geq 1} P(X \geq k)$. These crack “expected number of trials” MCQs in seconds.
- KEY. Linearity is unconditional:** $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$ requires *no independence whatsoever*. Half of all “expected number of matches/records” items are solved by indicators + linearity: e.g. expected number of fixed points of a random permutation = 1, for every n .
- Products: $X \perp Y \Rightarrow \mathbb{E}(XY) = \mathbb{E}X \mathbb{E}Y$. **TRAP.** Converse false (uncorrelated $\neq$ independent). And $\mathbb{E}(X/Y) \neq \mathbb{E}X/\mathbb{E}Y$; $\mathbb{E}(1/X) \neq 1/\mathbb{E}X$ — Jensen gives $\mathbb{E}(1/X) \geq 1/\mathbb{E}X$ for positive X .

7. Jensen: g convex $\Rightarrow \mathbb{E}g(X) \geq g(\mathbb{E}X)$ (reverse for concave; equality iff g affine on the support or X degenerate). Consequences: $\mathbb{E}X^2 \geq (\mathbb{E}X)^2$ (i.e. $\text{Var} \geq 0$); $\mathbb{E}e^{tX} \geq e^{t\mathbb{E}X}$; $\mathbb{E} \ln X \leq \ln \mathbb{E}X$.
8. Cauchy–Schwarz: $(\mathbb{E}XY)^2 \leq \mathbb{E}X^2 \mathbb{E}Y^2$; equality iff linear dependence a.s. This is also why $|\rho| \leq 1$ and $|\rho| = 1$ iff $Y = aX + b$.
9. Variance identities: $\text{Var}(aX + b) = a^2 \text{Var}X$ (**TRAP.** the “+b” vanishes, a is squared — so $\text{Var}(-X) = \text{Var}X$); $\text{Var}(X \pm Y) = \text{Var}X + \text{Var}Y \pm 2\text{Cov}(X, Y)$ — variances *add* for the difference of uncorrelated variables. Cov is bilinear; $\text{Cov}(X, c) = 0$; correlation is invariant under positive affine maps and flips sign under negative scaling.

8.2 Conditional expectation: the seven laws

1. $\mathbb{E}[X | Y]$ is a **random variable** (a function of Y), not a number. MCQs that treat it as a constant are auto-false.
2. **Tower / total expectation:** $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}X$. Total variance: $\text{Var}X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$ — both terms ≥ 0 , hence conditioning *never increases* expected residual variance: $\mathbb{E}[\text{Var}(X | Y)] \leq \text{Var}X$.
3. Taking out what is known: $\mathbb{E}[g(Y)X | Y] = g(Y) \mathbb{E}[X | Y]$.
4. Independence: $X \perp Y \Rightarrow \mathbb{E}[X | Y] = \mathbb{E}X$. **TRAP.** Converse false: $\mathbb{E}[X | Y] = \mathbb{E}X$ (*mean independence*) does not force independence (e.g. $X = YZ$ with symmetric Z). Hierarchy: independence $\Rightarrow$ mean independence $\Rightarrow$ uncorrelated; both arrows strict.
5. Best predictor: $\mathbb{E}[X | Y]$ minimises $\mathbb{E}(X - g(Y))^2$ over all g — the regression function. The best *linear* predictor is generally different (they coincide under joint normality — another “jointly” trap).
6. Symmetry trick: for iid $X_1, \dots, X_n$ and $S = \sum X_i$: $\mathbb{E}[X_1 | S] = S/n$. Instant answer to a recurring exam item.
7. Standard-model conditionals to memorise: Poisson split $X | X + Y \sim \text{Binomial}$; multinomial cell given another cell $\sim \text{Binomial}$; bivariate normal: $\mathbb{E}[Y | X] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$ (linear!) with constant conditional variance $\sigma_Y^2(1 - \rho^2)$.

9 Generating Functions: CF, MGF, PGF — Inversion, Uniqueness, Continuity

9.1 Characteristic functions: what is always true

$$\varphi_X(t) = \mathbb{E}e^{itX}.$$

1. **KEY.** Exists for **every** distribution (since $|e^{itX}| = 1$) — the decisive advantage over the MGF. Properties: $\varphi(0) = 1$; $|\varphi(t)| \leq 1$; uniformly continuous on $\mathbb{R}$; Hermitian $\varphi(-t) = \overline{\varphi(t)}$; $\varphi_{aX+b}(t) = e^{ibt} \varphi_X(at)$.
2. φ is real-valued $\iff X$ is symmetric about 0. **MCQ angle:** “ φ real $\Rightarrow$ mean 0” — False (Cauchy: real CF, no mean).
3. Independence: $\varphi_{X+Y} = \varphi_X \varphi_Y$. **TRAP.** The converse is **False:** for X standard Cauchy and $Y = X$ (maximally dependent), $\varphi_{X+Y}(t) = e^{-2|t|} = \varphi_X(t) \varphi_Y(t)$. Factorisation of the CF does not certify independence.

4. **TRAP.** Not every nice function is a CF: $|\varphi(t)| \leq 1$ with $\varphi(0) = 1$ is necessary, not sufficient (Bochner's theorem gives the full criterion: non-negative definiteness). E.g. e^{-t^4} is *not* a CF (a genuine top-1% item: if it were, the fourth-moment/kurtosis relations force a degenerate-normal contradiction).
5. Moments: if $\mathbb{E}|X|^n < \infty$ then $\varphi^{(n)}(0) = i^n \mathbb{E}X^n$; in particular $\varphi'(0) = i\mathbb{E}X$, $\varphi''(0) = -\mathbb{E}X^2$. **TRAP.** Differentiability of φ at 0 does *not* guarantee the mean exists (one-way street for odd orders; existence of $\varphi^{(2k)}(0)$ *does* force $\mathbb{E}X^{2k} < \infty$ — even orders are special).
6. Landmark CFs: $N(0,1)$: $e^{-t^2/2}$ (self-conjugate shape); Cauchy: $e^{-|t|}$; Poisson: $e^{\lambda(e^{it}-1)}$; $U(-1,1)$: $\frac{\sin t}{t}$; Laplace(0,1): $\frac{1}{1+t^2}$; degenerate at c : e^{ict} .

9.2 The three big theorems (statement precision wins the mark)

1. **Uniqueness:** $\varphi_X = \varphi_Y$ (for all t) $\iff X \stackrel{d}{=} Y$. CFs are fingerprints. **TRAP.** Agreement on an interval is *not* enough in general.
2. **Inversion:** for continuity points $a < b$: $P(a < X \leq b) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-ita} - e^{-itb}}{it} \varphi(t) dt$; if $\int |\varphi| < \infty$, X has the bounded continuous density $f(x) = \frac{1}{2\pi} \int e^{-itx} \varphi(t) dt$. **MCQ angle:** “integrable CF $\Rightarrow$ density exists” — True; converse False.
3. **Lévy continuity:** $X_n \xrightarrow{d} X \iff \varphi_n(t) \rightarrow \varphi(t)$ for every t with φ a CF; equivalently the pointwise limit is **continuous at 0**. **TRAP.** The continuity-at-0 clause is the exam's favourite: $X_n \sim N(0,n)$ gives $\varphi_n(t) = e^{-nt^2/2} \rightarrow \mathbf{1}\{t=0\}$, discontinuous at 0 — and indeed the mass escapes to $\pm\infty$, no limiting distribution. This single example is worth a full mark every time it appears.

9.3 MGF and PGF: existence fine print

1. MGF $M(t) = \mathbb{E}e^{tX}$ may fail to exist for all $t \neq 0$ (Cauchy) or all $t > 0$ (lognormal). If M exists on a neighbourhood $(-\delta, \delta)$: all moments are finite, $M^{(k)}(0) = \mathbb{E}X^k$, and M **determines the distribution**.
2. **TRAP.** “All moments finite $\Rightarrow$ distribution determined by moments” is **False** (lognormal). “MGF exists near 0 $\Rightarrow$ determined” is True. Keep the two statements separated in memory; the exam will put them side by side.
3. $M_{X+Y} = M_X M_Y$ under independence (same caveat as CFs on the converse); $M_{aX+b}(t) = e^{bt} M_X(at)$; $M(0) = 1$ always; M is convex; $M(t) \geq e^{t\mathbb{E}X}$ (Jensen).
4. PGF (non-negative integer variables): $G(s) = \mathbb{E}s^X = \sum p_k s^k$, radius ≥ 1 ; $G(1) = 1$; $p_k = G^{(k)}(0)/k!$ (*probabilities* from derivatives at 0); factorial moments from derivatives at 1: $G'(1) = \mathbb{E}X$, $G''(1) = \mathbb{E}X(X-1)$, so $\text{Var}X = G''(1) + G'(1) - [G'(1)]^2$ — this formula is asked directly.
5. $G_{X+Y} = G_X G_Y$ (independent); random sums: $\mathbb{E}s^{S_N} = G_N(G_X(s))$ (compound/branching PGF) with $\mathbb{E}S_N = \mathbb{E}N \mathbb{E}X$ (Wald) — top-1% but appears in “number of offspring”-style items.

10 Zero–One Laws and the Classical Inequalities

10.1 Borel–Cantelli and Borel's 0–1 law

1. **BC–I:** $\sum P(A_n) < \infty \Rightarrow P(A_n \text{ i.o.}) = 0$. **KEY.** Needs *no* independence — planted MCQs claim it does.

2. **BC-II:** if the A_n are **independent** and $\sum P(A_n) = \infty$, then $P(A_n \text{ i.o.}) = 1$. **TRAP.** Independence here is essential: take $A_n = A$ with $P(A) = \frac{1}{2}$ for all n ; the sum diverges but $P(A_n \text{ i.o.}) = \frac{1}{2}$.
3. **Borel's 0–1 law:** for *independent* events, $P(A_n \text{ i.o.})$ is 0 or 1 according as $\sum P(A_n)$ converges or diverges — no middle value possible. Infinite-monkey corollary: any fixed finite pattern in iid typing occurs infinitely often with probability 1.
4. i.o. notation: $\{A_n \text{ i.o.}\} = \limsup A_n = \bigcap_n \bigcup_{k \geq n} A_k$; $\liminf A_n = \{A_n \text{ eventually always}\}$; $\liminf A_n \subseteq \limsup A_n$; $P(\liminf A_n) \leq \liminf P(A_n) \leq \limsup P(A_n) \leq P(\limsup A_n)$ (Fatou for sets) — the inequality chain itself is an MCQ.

10.2 Kolmogorov's 0–1 law

1. For an **independent** sequence $X_1, X_2, \dots$, every *tail event* (unchanged by altering finitely many X_i 's) has probability 0 or 1.
2. Tail events (probability forced to $\{0, 1\}$): $\{\sum X_n \text{ converges}\}$, $\{\lim X_n \text{ exists}\}$, $\{\limsup X_n > c\}$, $\{X_n \in B \text{ i.o.}\}$, $\{\bar{X}_n \text{ converges}\}$. Non-tail (trap options): $\{X_1 > 0\}$, $\{\sum X_n > 0\}$, $\{S_n = 0 \text{ i.o.}\}$ — these depend on early terms.
3. **TRAP.** The law says the probability *is* 0 or 1; it does not tell you *which*. Determining which one is the hard analytic work (e.g. via Borel–Cantelli or the three-series theorem).
4. Consequences: $\limsup X_n$, $\limsup S_n/n$ etc. are a.s. *constants* (possibly $\pm\infty$) for independent sequences.

10.3 Tchebycheff, Markov, Kolmogorov: exact hypotheses, exact bounds

1. **Markov:** $X \geq 0$, $a > 0$: $P(X \geq a) \leq \mathbb{E}X/a$. **TRAP.** Non-negativity is essential; the bound is useless for $a \leq \mathbb{E}X$ (bound ≥ 1).
2. **Tchebycheff:** $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, equivalently $P(|X - \mu| \geq \varepsilon) \leq \sigma^2/\varepsilon^2$. Needs finite variance, **no** distributional shape. **MCQ angle:** Complement form: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$ (at least 75% within 2σ , 88.9% within 3σ — compare with 95.45%, 99.73% for the normal: “universal vs normal” pairs are repeatedly asked).
3. **TRAP.** Tchebycheff is **sharp**: $X = \pm k\sigma$ w.p. $\frac{1}{2k^2}$ each and 0 w.p. $1 - \frac{1}{k^2}$ attains equality — so “the inequality can be improved without further assumptions” is False. One-sided (Cantelli): $P(X - \mu \geq \varepsilon) \leq \frac{\sigma^2}{\sigma^2 + \varepsilon^2}$ — strictly better on one side.
4. **Kolmogorov's inequality:** independent, mean-zero, finite-variance X_i , $S_k = \sum_{i \leq k} X_i$:

$$P\left(\max_{1 \leq k \leq n} |S_k| \geq \varepsilon\right) \leq \frac{\text{Var}(S_n)}{\varepsilon^2}.$$

KEY. It bounds the *maximum* of all partial sums by the same quantity Tchebycheff uses for S_n alone — “Kolmogorov strengthens Tchebycheff” is the expected true statement; it is the engine of the SLLN proof.

11 Laws of Large Numbers and Central Limit Theorems

11.1 WLLN vs SLLN: hypothesis bookkeeping (pure MCQ fuel)

1. **Tchebycheff/Markov WLLN** (not nec. identically distributed): pairwise uncorrelated with $\text{Var}(S_n)/n^2 \rightarrow 0$ (e.g. uniformly bounded variances) $\Rightarrow \bar{X}_n - \mathbb{E}\bar{X}_n \xrightarrow{P} 0$. Needs variances.

2. **Khinchin’s WLLN:** iid with finite mean $\Rightarrow \bar{X}_n \xrightarrow{P} \mu$. **KEY. No variance assumption.** “WLLN requires finite variance” — **False** (for iid); this is among the most re-asked discriminator items.
3. **Kolmogorov’s SLLN (iid):** $\bar{X}_n \xrightarrow{\text{a.s.}} \mu \iff \mathbb{E}|X_1| < \infty$ (and then $\mu = \mathbb{E}X_1$). Finite mean is necessary *and* sufficient.
4. **Kolmogorov’s SLLN (independent, non-iid):** $\sum_n \text{Var}(X_n)/n^2 < \infty \Rightarrow \frac{S_n - \mathbb{E}S_n}{n} \xrightarrow{\text{a.s.}} 0$ (Kolmogorov’s criterion). Note the n^2 in the denominator — MCQs swap it with n .
5. **Cauchy kills everything:** iid Cauchy has $\bar{X}_n \sim \text{Cauchy}$ for all n : no WLLN, no SLLN, no CLT. The st. Petersburg variable (infinite mean) gives $\bar{X}_n \rightarrow \infty$. “LLN holds for every iid sequence” — False; finite mean is the gatekeeper.
6. SLLN $\Rightarrow$ WLLN (a.s. implies in probability). A sequence can obey the WLLN and violate the SLLN (independent, suitably weighted examples) — direction matters.
7. Bernoulli LLN: relative frequency $\xrightarrow{P} p$ — the formal bridge between the frequency interpretation and the axioms; also $\xrightarrow{\text{a.s.}}$ by Borel’s SLLN (1909), historically the first strong law.

11.2 CLT: the three named versions

1. **Lindeberg–Lévy (iid):** iid, $\mathbb{E}X = \mu$, $\text{Var}X = \sigma^2 \in (0, \infty)$:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Both moments finite is the *only* distributional assumption — no symmetry, no continuity. **TRAP.** Finite variance is essential: iid Cauchy fails; iid Pareto with infinite variance converges to *stable*, non-normal, laws.

2. **de Moivre–Laplace:** the binomial special case, $\frac{X - np}{\sqrt{npq}} \xrightarrow{d} N(0, 1)$; with continuity correction $P(X \leq k) \approx \Phi\left(\frac{k + 0.5 - np}{\sqrt{npq}}\right)$ — expect a numeric MCQ where the 0.5 decides between two options.
3. **Lyapunov (independent, non-iid):** if for some $\delta > 0$ $\frac{\sum_i \mathbb{E}|X_i - \mu_i|^{2+\delta}}{s_n^{2+\delta}} \rightarrow 0$ (with $s_n^2 = \sum \text{Var}X_i$), then $\frac{S_n - \mathbb{E}S_n}{s_n} \xrightarrow{d} N(0, 1)$. **KEY.** Lyapunov’s condition is *sufficient, not necessary*; the sharp iff-type condition (given uniform negligibility) is **Lindeberg–Feller** — knowing which name carries the “necessary and sufficient” label is a top-1% discriminator.
4. CLT vs LLN division of labour: LLN says $\bar{X}_n \rightarrow \mu$; CLT says the error is of exact order $n^{-1/2}$ with a universal Gaussian shape: $\bar{X}_n \approx N(\mu, \sigma^2/n)$. Berry–Esseen quantifies the speed: $\sup_x |F_n(x) - \Phi(x)| \leq C\rho/(\sigma^3\sqrt{n})$ (order $n^{-1/2}$, needs a third moment).
5. **TRAP.** The CLT is about the *standardised sum*: S_n itself diverges, and $\bar{X}_n$ collapses to a constant; neither is asymptotically normal without the $\sqrt{n}$ scaling — sloppy MCQ stems exploit this.
6. Proof-sketch fact examiners quote: CF method — $\varphi_{S_n/\sqrt{n}}(t) = \left(1 - \frac{t^2}{2n} + o(1/n)\right)^n \rightarrow e^{-t^2/2}$, then Lévy continuity. Know this two-line skeleton; “which theorem finishes the CF proof of the CLT?” has appeared in several exams’ orbit.

12 The 2027 Expected-Question Bank: 60 Trap Statements with Verdicts

KEY. Each statement below is written exactly as an MCQ option would be. Decide True/False *before* reading the verdict in brackets. Target: 58 + /60 with reasons. Anything you miss goes into the error ledger (§14).

12.1 Foundations and conditioning

1. If $P(A) = 0$ then $A = \emptyset$. **[False]** null $\neq$ empty; $U(0, 1)$ hits any point w.p. 0.
2. Mutually exclusive events with positive probabilities can be independent. **[False]** $P(A \cap B) = 0 < P(A)P(B)$.
3. If A is independent of itself, $P(A) \in \{0, 1\}$. **[True]** $P(A) = P(A)^2$.
4. Pairwise independence of A, B, C implies $P(A \cap B \cap C) = P(A)P(B)P(C)$. **[False]** Bernstein's two-coin example.
5. A σ -field on a finite set can contain exactly 6 events. **[False]** cardinality is 2^k .
6. $P(A | B) + P(A^c | B) = 1$ whenever $P(B) > 0$. **[True]** $P(\cdot | B)$ is a probability measure.
7. $P(A | B) + P(A | B^c) = 1$ in general. **[False]** no such identity.
8. If $P(A | C) > P(B | C)$ and $P(A | C^c) > P(B | C^c)$, then $P(A) > P(B)$. **[True]** here — mixture of strict inequalities with the *same* weights $P(C), P(C^c)$; contrast with Simpson's paradox, where the comparisons $P(A | C)$ vs $P(B | C)$ use *different* conditioning weights for A-type and B-type groups (two-treatment, two-hospital data). Read stems letter by letter.
9. In the Monty Hall game with a knowing host, switching wins with probability $2/3$. **[True]** and with an ignorant host who happens to reveal a goat, $1/2$.
10. A positive test from a 99%-accurate test for a 1-in-1000 disease implies $P(\text{disease}) \approx 0.99$. **[False]** ≈ 0.09 ; base rates rule.
11. "Two children, at least one boy": $P(\text{both boys}) = 1/2$. **[False]** $1/3$; it becomes $1/2$ if the *identified* (e.g. elder) child is a boy.
12. In sampling without replacement, the probability the second draw is red equals the initial red proportion. **[True]** symmetry/exchangeability.

12.2 Random variables, CDFs, distributions

1. Every CDF is continuous from the left. **[False]** right-continuity is the axiom; left-limits give $P(X < x)$.
2. A CDF can have uncountably many discontinuities. **[False]** jumps are countable.
3. If a CDF is continuous everywhere, the random variable has a density. **[False]** Cantor distribution.
4. A valid pdf never exceeds 1. **[False]** $U(0, \frac{1}{2})$: $f \equiv 2$; $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ unbounded.
5. If F and G are CDFs, FG is a CDF. **[True]** law of $\max(X, Y)$ under independence.
6. If F is a CDF, $1 - (1 - F)^3$ is a CDF. **[True]** minimum of 3 iid copies.

7. X and $-X$ having the same distribution implies $X = 0$ a.s. **[False]** any symmetric law; equal in distribution $\neq$ equal.
8. For continuous F , $F(X) \sim U(0, 1)$. **[True]** probability integral transform.
9. A binomial variable can have mean 6 and variance 8. **[False]** needs $q = 4/3 > 1$; variance $<$ mean always.
10. A Poisson variable with integer mean λ has two modes. **[True]** $\lambda - 1$ and λ .
11. If X, Y are independent Poisson, $X - Y$ is Poisson. **[False]** negative values; Skellam.
12. If $X \perp Y$ and $X + Y$ is Poisson, both X, Y are Poisson. **[True]** Raikov's theorem. (Normal analogue: Cramér's theorem — also True.)
13. The exponential distribution satisfies $P(X > s + t \mid X > s) = P(X > t)$, and it is the unique continuous law on $(0, \infty)$ doing so. **[True]** memorylessness characterisation.
14. For exponential X , $P(X > \mathbb{E}X) = \frac{1}{2}$. **[False]** $e^{-1} \approx 0.368$; median $<$ mean.
15. The sum of two independent $U(0, 1)$ variables is $U(0, 2)$. **[False]** triangular.
16. The sum of two independent exponentials with the same rate is exponential. **[False]** Gamma(2, λ); the *minimum* is exponential.
17. If X, Y are normal, $X + Y$ is normal. **[False]** needs independence or joint normality; sign-flip counterexample has an atom at 0.
18. Uncorrelated jointly normal variables are independent. **[True]** the “jointly” does all the work.
19. The Cauchy distribution has mean 0 by symmetry. **[False]** the mean does not exist; use the median.
20. Some distributions possess every moment yet have no MGF for $t > 0$. **[True]** lognormal.
21. Two different distributions can share all moments. **[True]** lognormal's moment-indeterminacy.
22. Hypergeometric sampling gives a larger variance than binomial with the same n and $p = K/N$. **[False]** smaller: FPC $\frac{N-n}{N-1} < 1$.
23. In a multinomial, $\text{Cov}(X_i, X_j) > 0$ can occur. **[False]** always $-np_i p_j < 0$.
24. For the Laplace(μ, b) law, the variance is b^2 . **[False]** $2b^2$; and the MLE of μ is the sample median.
25. For a negative binomial count of failures, variance exceeds mean. **[True]** overdispersion signature (binomial under-, Poisson equi-).
26. Beta(1, 1) is the uniform law on $(0, 1)$. **[True]** and $U_{(k)} \sim \text{Beta}(k, n + 1 - k)$.
27. The Gamma(α, λ) MGF exists for all real t . **[False]** only $t < \lambda$.
28. For lognormal: mean $<$ median. **[False]** mean $e^{\mu + \sigma^2/2} >$ median $e^\mu >$ mode $e^{\mu - \sigma^2}$.

12.3 Convergence, expectation, generating functions, limit theorems

1. Convergence in probability implies almost-sure convergence. **[False]** independent indicators with $p_n = 1/n$; a subsequence does converge a.s.
2. Convergence in distribution to a constant implies convergence in probability. **[True]** the one upgrade that is free.
3. $X_n \xrightarrow{d} X$ requires $F_n(x) \rightarrow F(x)$ for every real x . **[False]** only at continuity points of F ; $X_n = 1/n$ vs $x = 0$.
4. If $X_n \xrightarrow{\text{a.s.}} X$ then $\mathbb{E}X_n \rightarrow \mathbb{E}X$. **[False]** tall-spike example; needs uniform integrability/domination.
5. If $X_n \rightarrow X$ in mean square, then $X_n \xrightarrow{P} X$. **[True]** Markov/Tchebycheff on $|X_n - X|^2$.
6. $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ imply $X_n + Y_n \xrightarrow{d} X + Y$. **[False]** joint law unspecified; true under Slutsky (Y constant) or independence.
7. $\mathbb{E}(X + Y) = \mathbb{E}X + \mathbb{E}Y$ requires independence. **[False]** linearity is unconditional — the most profitable “True/False” in the paper.
8. $\mathbb{E}(XY) = \mathbb{E}X\mathbb{E}Y$ implies $X \perp Y$. **[False]** $X \sim N(0, 1)$, $Y = X^2$.
9. $\mathbb{E}[X | Y]$ is a random variable and $\mathbb{E}\{\mathbb{E}[X | Y]\} = \mathbb{E}X$. **[True]** tower property.
10. $\mathbb{E}[X | Y] = \mathbb{E}X$ implies $X \perp Y$. **[False]** mean-independence is weaker.
11. $\text{Var}X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$. **[True]** total variance; both pieces ≥ 0 .
12. For iid $X_1, \dots, X_n$, $\mathbb{E}[X_1 | \sum_i X_i] = \bar{X}_n \cdot n/n = \sum_i X_i/n$. **[True]** symmetry.
13. A characteristic function may fail to exist for heavy-tailed laws. **[False]** CFs always exist; MGFs may not.
14. $\varphi_{X+Y} = \varphi_X \varphi_Y$ implies $X \perp Y$. **[False]** Cauchy $Y = X$ example.
15. A real-valued CF implies the distribution is symmetric about 0. **[True]** and conversely.
16. If $\varphi_n(t) \rightarrow e^{-t^2/2}$ pointwise, then $X_n \xrightarrow{d} N(0, 1)$. **[True]** Lévy continuity (limit is a CF, continuous at 0).
17. If $\varphi_n(t) \rightarrow L(t)$ pointwise, X_n always converges in distribution. **[False]** $N(0, n)$: $L = \mathbf{1}\{t = 0\}$ discontinuous at 0, mass escapes.
18. $\sum P(A_n) < \infty$ implies $P(A_n \text{ i.o.}) = 0$ only for independent events. **[False]** BC–I needs no independence; independence belongs to BC–II.
19. For independent events with $\sum P(A_n) = \infty$, $P(A_n \text{ i.o.}) = 1$. **[True]** Borel’s 0–1 law.
20. For an independent sequence, $\{\lim_n \bar{X}_n \text{ exists}\}$ has probability 0 or 1. **[True]** tail event; Kolmogorov’s 0–1 law.
21. Tchebycheff’s inequality can be sharpened without further assumptions. **[False]** three-point law attains it.
22. Kolmogorov’s inequality bounds $P(\max_{k \leq n} |S_k| \geq \varepsilon)$ by $\text{Var}(S_n)/\varepsilon^2$. **[True]** maximal strengthening of Tchebycheff.
23. The WLLN for iid sequences requires finite variance. **[False]** Khinchin needs only finite mean.

24. The SLLN holds for iid Cauchy variables. **[False]** no finite mean; $\bar{X}_n \sim \text{Cauchy}$ forever.
25. Lyapunov’s condition is necessary for the CLT of independent arrays. **[False]** sufficient only; Lindeberg–Feller carries the iff (under negligibility).
26. By CLT, S_n itself converges in distribution to a normal law. **[False]** only the standardised $(S_n - n\mu)/(\sigma\sqrt{n})$ does.

KEY. Scoring guide: 58–60: exam-ready on theory traps; 52–57: re-read the matching sections and retest in 48 hours; below 52: restart the compendium with the counterexamples first.

13 The Mistakes That Waste 80% of Beginners’ Time

1. **Reading theory before PYQs.** You cannot recognise a trap you have never been cut by. Invert: questions first, theory as the debrief.
2. **Deriving in the exam what should be a lookup.** Modes, memorylessness, closure-under-addition, mean/variance of all 16 families, the convergence hierarchy: these must be reflexes (<10 seconds), not derivations. Every derivation in the hall is 2 minutes stolen from 3 other questions.
3. **Learning distributions as isolated formulas** instead of as a *network* (limit lattice, closure table, characterisations). Matching questions are set on the network, not the formulas.
4. **Ignoring parametrisation conventions.** Geometric (from 0 or 1), negative binomial (trials or failures), gamma (rate or scale), exponential (λ or mean θ): more marks die here than on any genuinely hard idea.
5. **Believing symmetry authorises expectations.** The Cauchy mean does not exist. Full stop.
6. **Treating a.s. convergence and mean-square convergence as ordered.** They are not comparable; keep the two counterexamples loaded.
7. **Confusing pairwise with mutual independence,** and uncorrelatedness with independence. Three different notions, strictly ordered only one way.
8. **Forgetting the support.** Non-rectangular support kills independence regardless of the formula; convolution limits come from supports; conditional densities exist only where the marginal is positive.
9. **Skipping the “jointly” in jointly normal.** Marginal normality is almost never enough for anything.
10. **Over-practising integration, under-practising verdicts.** ISS objective papers reward 90-second true/false discrimination, not 15-minute integrals. Allocate practice in that proportion.
11. **No error ledger.** Unlogged mistakes recur with probability ≈ 1 (a very strong law). The ledger, revised weekly, is the highest-ROI hour of the week.
12. **Leaving eliminate-one questions blank.** Expected value of a guess after one elimination is positive ($\approx +0.28$ marks). Blind blanks are a scoring error, not caution.
13. **Mock-hoarding.** Twenty analysed mocks beat sixty unanalysed ones. The analysis (trap-class tagging) is the practice.
14. **Studying limit theorems as folklore** (“everything becomes normal”) instead of as precise statements with hypotheses. The marks sit in the hypotheses: iid?, finite mean?, finite variance?, independence?, standardisation?

14 The Aggressive 2027 Practice Plan

KEY. Assumption: ISS 2027 written exam around mid-2027 (UPSC typically notifies in February and examines in June); you start now (July 2026): ≈ 46 weeks. This plan covers the Probability block of Statistics-I; run parallel tracks for the other blocks with the same architecture.

14.1 Phase 0 — Calibration (Weeks 1–2)

1. Attempt every probability question from ISS Statistics-I PYQs 2020–2025, *untimed*, *closed book*. Record: correct / wrong / lucky / blank.
2. Tag every error with a trap-class from this compendium (e.g. “support”, “parametrisation”, “hierarchy”, “hypothesis of theorem”). This tagging *is* the diagnosis; it tells you which sections here to over-weight.
3. Set up the error ledger (one line per error: date, source, trap-class, the one-sentence reason, retest date +7d and +30d).

14.2 Phase 1 — Trap-first mastery (Weeks 3–14; the engine room)

Six blocks, two weeks each, in this order (dependency-sorted): (1) Foundations + conditioning/Bayes; (2) RVs + CDFs + univariate transformations; (3) The 16 distributions (one family per morning, cross-tables at week end); (4) Random vectors + Jacobians + order statistics; (5) Expectation + conditional expectation + CF/MGF/PGF; (6) Convergence + inequalities + 0–1 laws + LLN/CLT.

Daily rhythm (2.5–3 h):

1. 30 min: read the matching compendium section *actively* (cover the verdict, predict, check).
2. 75 min: 25–35 MCQs on that block. Sources in priority order: ISS PYQs (2016–2025), IIT JAM (MS) PYQs, GATE (ST) PYQs, CSIR–NET Statistics Part B — these share the ISS question-setting culture; standard drill books (Gupta–Kapoor for volume; Rohatgi–Saleh and Casella–Berger exercises for the top-1% edge) fill gaps.
3. 20 min: ledger entries + flashcards (every **TRAP**. line in this file is one card).
4. Weekly: one 25-question mixed timed set (30 min) across all blocks opened so far + the 60-item bank of §12 as a monthly full retest.

14.3 Phase 2 — Mixed-speed training (Weeks 15–30)

1. Alternate days: (a) 40-question mixed probability sprints at 75 s/question; (b) full-syllabus Statistics-I sectionals where probability is embedded among other topics — discrimination under interference is the real exam skill.
2. Introduce the **90-second protocol** per MCQ: 10 s classify the trap-class $\rightarrow$ 40 s apply the known verdict/counterexample $\rightarrow$ 20 s eliminate $\rightarrow$ decide (answer if $\geq$ one option eliminated) $\rightarrow$ 20 s buffer. Practise the protocol until it is automatic.
3. Ledger metric to track weekly: *repeat-error rate* (same trap-class twice). Target $< 10\%$ by week 24. Accuracy target on sprints: 85% by week 20, 90% by week 30.

14.4 Phase 3 — Exam simulation (Weeks 31–42)

1. Two full Statistics–I mocks weekly (80 Q / 120 min) under strict conditions, each followed the *same day* by a 90-minute autopsy: every error tagged, every guess audited (was the elimination real?), negative-marking discipline scored.
2. Rebuild speed reserves: first pass ≤ 90 min answering all sure questions; second pass for the eliminate-and-decide pool; never a third pass.
3. Weeks 39–42: taper new material to zero; only ledger, flashcards, the 60-item bank, and the cross-family tables of §5.16.

14.5 Phase 4 — Peak (final 4 weeks before the paper)

1. Daily 30-minute recall of: closure table, limit lattice, mean/variance table, convergence hierarchy + four counterexamples, the three CF theorems, hypothesis tables of the LLNs/CLTs, and the paradox verdicts.
2. Three final mocks, spaced 4–5 days, last one 7 days out. No mock in the final week; the final week belongs to the ledger only.
3. Sleep is a study technique: memory consolidation of exactly this kind of discrete fact-lattice happens offline. Protect it.

14.6 Non-negotiables (the whole plan in five lines)

1. PYQs before theory; theory as debrief.
2. Every mistake becomes a ledger line with a trap-class and two retest dates.
3. Counterexamples are carried as flashcards until they answer themselves in <10 s.
4. Timed practice at 75–90 s/question from week 15 onward, with the guessing arithmetic applied ruthlessly.
5. The 60-item bank of §12 is retaken monthly; 58+ or the block reopens.

Final word from the faculty desk. The syllabus paragraph you are chasing — from Kolmogorov’s axioms to the central limit theorem — is a single connected story: measures $\rightarrow$ variables $\rightarrow$ distributions $\rightarrow$ transformations $\rightarrow$ generating functions $\rightarrow$ convergence $\rightarrow$ limit laws. The exam samples the story at its joints. Own the joints (every **TRAP** in this file), and the 2027 paper will read like a list of questions you have already answered.